

Differentials and Partial Derivatives

Ex8.1

Question 1.

Let $f(x) = \sqrt[3]{x}$. Find the linear approximation at $x=27$. Use the linear approximation to $\sqrt[3]{27.2}$.

Solution:

$$f(x) = \sqrt[3]{x} = x^{1/3} \quad f'(x) = \frac{1}{3}x^{-2/3} \quad f(x_0) = f(27) = (27)^{1/3} = 3 \quad x_0 = 27 \quad \Delta x = 0.2 \quad x = 27.2$$

$$f'(x_0) = f'(27) = \frac{1}{3}(27)^{-2/3} = \frac{1}{3}(3)^{-2} = \frac{1}{3}(1/9) = 1/27$$

The required linear approximation $L(x) = f(x_0) + f'(x_0)(x - x_0)$

$$L(x) = 3 + \frac{1}{27}(x - 27) \quad L(27.2) = 3 + \frac{1}{27}(27.2 - 27) = 3 + \frac{1}{27}(0.2) = 3 + 0.0074 = 3.0074 \text{ (approximately)}$$

Question 2.

Use the linear approximation to find approximate values of

(i) $(123)^{2/3}$

(ii) $\sqrt[4]{15}$

(iii) $\sqrt[3]{26}$

Solution:

(i) $(123)^{2/3} = (125 - 2)^{2/3}$

we have,

$$x = 123 \quad x_0 = 125 \quad \Delta x = -2$$

$$\text{Let } f(x) = x^{2/3}$$

$$f(x) = \frac{2}{3}x^{-1/3} \quad f(x_0) = f(125) = (125)^{2/3} = 25 \quad f'(x_0) = \frac{2}{3}(125)^{-1/3} = \frac{2}{3}(5)^{-1} = \frac{2}{15}$$

$$x = 123 \quad x_0 = 125 \quad \Delta x = -2$$

The required linear approximation $L(x) = f(x_0) + f'(x_0)(x - x_0)$

$$L(x) = 25 + \frac{2}{15}(x - 125) \quad L(123) = 25 + \frac{2}{15}(123 - 125) = 25 + \frac{2}{15}(-2) = 25 - \frac{4}{15} = 25 - 0.2667 = 24.7333 \text{ (approximately)}$$

(ii) $\sqrt[4]{15}$ consider $(15)^{1/4} = (16 - 1)^{1/4}$

we have,

$$x = 15 \quad x_0 = 16 \quad \Delta x = -1$$

$$\text{Let } f(x) = x^{1/4} \text{ we have,}$$

$$x = 15 \quad x_0 = 16 \quad \Delta x = -1$$

$$f(x) = \frac{1}{4}x^{-3/4} \quad f(x_0) = f(16) = (16)^{1/4} = 2$$

$$f'(x_0) = f'(16) = \frac{1}{4}(16)^{-3/4} = \frac{1}{4}(2)^{-3} = \frac{1}{32}$$

The required linear approximation $L(x) = f(x_0) + f'(x_0)(x - x_0)$

$$L(x) = 2 + \frac{1}{32}(x - 16) \quad L(15) = 2 + \frac{1}{32}(15 - 16) = 2 - \frac{1}{32} = 2 - 0.03125 = 1.96875 \text{ (approximately)}$$

(iii)

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we have,

$$x = 26 \quad x_0 = 27 \quad \Delta x = -1$$

$$\sqrt[3]{26} = (26)^{1/3} = (27 - 1)^{1/3} \text{ consider } f(x) = x^{1/3}$$

$$f(x) = \frac{1}{3}x^{-2/3} \quad x = 26 \quad x_0 = 27 \quad \Delta x = -1 \quad f(x_0) = f(27) = (27)^{1/3} = 3 \quad f'(x_0) = f'(27) = \frac{1}{3}(27)^{-2/3} = \frac{1}{3}(3)^{-2} = \frac{1}{27}$$

The required linear approximation $L(x) = f(x_0) + f'(x_0)(x - x_0)$

$L(x)=3+127(x-27)$
 $L(26)=3+127(26-27)=3+127(-1)=3-127=3-0.037=2.963$ (approximately)

Question 3.

Find a linear approximation for the following functions at the indicated points.

(i) $f(x)=x^3-5x+12, x_0=2$

(ii) $g(x)=\sqrt{x^2+9}, x_0=-4$

(iii) $h(x)=xx+1, x_0=1$

Solution:

(i) $f(x)=x^3-5x+12$
 $f'(x)=3x^2-5$
 $f(x_0)=f(2)=(2)^3-5(2)+12=8-10+12=10$
 $f'(x_0)=f'(2)=3(2)^2-5=12-5=7$

The required linear approximation $L(x)=f(x_0)+f'(x_0)(x-x_0)$
 $=10+7(x-2)=10+7x-14=7x-4$

(ii)
 $g(x)=\sqrt{x^2+9}; x_0=-4$
 $g'(x)=\frac{1}{2}\sqrt{x^2+9}^{-\frac{1}{2}}(2x)=\frac{x}{\sqrt{x^2+9}}$
 $g(x_0)=g(-4)=\sqrt{16+9}=\sqrt{25}=5$
 $g'(x_0)=\frac{-4}{\sqrt{25}}=-\frac{4}{5}$

The required linear approximation $L(x)=g(x_0)+g'(x_0)(x-x_0)$
 $=5-\frac{4}{5}(x+4)=5-\frac{4}{5}x-\frac{16}{5}=\frac{9}{5}-\frac{4}{5}x$

(iii)
 $h(x)=xx+1; x_0=1$
 $h'(x)=(x+1)(1)-x(1)=1$
 $h(x_0)=h(1)=1(1)+1=2$
 $h'(x_0)=h'(1)=1(1)+1=2$

The required linear approximation $L(x)=h(x_0)+h'(x_0)(x-x_0)$
 $=2+2(x-1)=2+x-2=x$

Question 4.

The radius of a circular plate is measured as 12.65 cm instead of the actual length 12.5 cm. find the following in calculating the area of the circular plate:

- (i) Absolute error
- (ii) Relative error
- (iii) Percentage error

Solution:

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We know that Area of the circular plate $A(r)=\pi r^2, A'(r)=2\pi r$

Change in Area $=A'(12.5)(0.15)=3.75\pi \text{ cm}^2$

Exact calculation of the change in Area $=A(12.65)-A(12.5)$
 $=160.0225\pi-156.25\pi=3.7725\pi \text{ cm}^2$

(i) Absolute error = Actual value - Approximate value
 $=3.7725\pi-3.75\pi$
 $=0.0225\pi \text{ cm}^2$

(ii)
 Relative error = $\frac{\text{Actual value} - \text{Approximate value}}{\text{Actual value}}$
 $=\frac{3.7725\pi-3.75\pi}{3.7725\pi}=\frac{0.0225\pi}{3.7725\pi}=0.006$

(iii)
 Percentage error = Relative error $\times 100=0.006 \times 100=0.6\%$

Question 5.

A sphere is made of ice having radius 10 cm. Its radius decreases from 10 cm to 9.8 cm. Find approximations for the following:

- (i) change in the volume

(ii) change in the surface area

Solution:

(i) We know that Volume of sphere

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$$v(r)=\frac{4}{3}\pi r^3 \quad v'(r)=4\pi(3r^2) \quad v'(r)=4\pi r^2$$

we have

$$r=10 \text{ cm}$$

Change in volume at $r=10$ is

$$=v'(r)[10-9.8]=4\pi(10)^2(0.2)=8\pi \text{ cm}^3$$

$\therefore$ Volume decreases by $80\pi \text{ cm}^3$

(ii) Surface area of the sphere

$$S(r)=4\pi r^2 \quad S'(r)=8\pi r$$

Change in surface area at $r=10$ is

$$=S'(r)[10-9.8]$$

$$=8\pi(10)(0.2)=16\pi \text{ cm}^2$$

$\therefore$ Surface Area decreases by $16\pi \text{ cm}^2$

Question 6.

The time T , taken for a complete oscillation of a single pendulum with length l , is given by the equation $T=2\pi\sqrt{\frac{l}{g}}$, where g is a constant. Find the approximate percentage error in the calculated value of T corresponding to an error of 2 percent in the value of l

Solution:

$$T=2\pi\sqrt{\frac{l}{g}} \quad \log T = \log 2\pi + \frac{1}{2}\log\left(\frac{l}{g}\right) \quad \log T = \log 2\pi + \frac{1}{2}[\log l - \log g] \quad \text{D.w.r to } l$$

$$\frac{1}{T} \frac{dT}{dl} = 0 + \frac{1}{2} \left[\frac{1}{l} - 0 \right] \quad \text{Percentage error d T} \quad \frac{dT}{T} = \frac{1}{2} \frac{dl}{l} \quad \Delta T \times 100 = \frac{1}{2} \Delta l \times 100 \quad (\text{But } dl=0.02l) = \frac{1}{2}(0.02l) \times 100 = 1\%$$

$$\text{error d T} \quad \frac{dT}{T} = \frac{1}{2} \frac{dl}{l} \quad \Delta T \times 100 = \frac{1}{2} \Delta l \times 100 \quad (\text{But } dl=0.02l) = \frac{1}{2}(0.02l) \times 100 = 1\%$$

Question 7.

Show that the percentage error in the n th root of a number is approximately $\frac{1}{n}$ times the percentage error in the number.

Solution:

Let x be the number

$$\therefore y=f(x)=(x)^{\frac{1}{n}} \quad \log y = \frac{1}{n} \log x \quad \text{D.w.r to } x \quad \frac{1}{y} \frac{dy}{dx} = \frac{1}{n} \frac{1}{x} \quad \frac{dy}{y} = \frac{1}{n} \frac{dx}{x}$$

Additional Questions Solved

Question 1.

Using differentials, find the approximate value of each of the following upto 3 places of decimal.

$(255)^{14}$ Solution:

$$\text{Let } y=x^{14} \quad \frac{dy}{dx}=14x^{13}$$

$$\text{Let } x=256 \text{ and } x+\Delta x=255$$

$$\Delta x=(x+\Delta x)-x=255-256=-1 \quad \Delta y=(x+\Delta x)^{14}-(x)^{14}=(255)^{14}-(256)^{14}=(255)^{14}-4$$

$$\therefore (255)^{14}=4+\Delta y$$

Since dy is approximately equal to Δy and is given by

$$dy=\frac{dy}{dx} \cdot \Delta x=14x^{13/4} \times (-1)=-14 \times (256)^{13/4}=-14 \times 64=-0.004 \therefore (255)^{14}=4-0.004=3.996$$

Thus approximate value of $(255)^{14}$ is 3.996.

Question 2.

Using differentials, find the approximate value of each of the following upto 3 places of decimal.

$$(401)_{12}$$

Solution:

$$\text{Let } y = x^{12} \Rightarrow dy = 12x^{11} dx$$

$$\text{Let } x = 400 \text{ and } x + \Delta x = 401$$

$$\Delta x = (x + \Delta x) - x = 401 - 400 = 1 \quad \Delta y = (x + \Delta x)^{12} - (x)^{12} \quad \Delta y = (x + \Delta x)^{12} -$$

$$(x)^{12} = (401)_{12} - (400)_{12} = (401)_{12} - 20$$

$$\therefore (401)_{12} = 20 + \Delta y$$

Since dy is approximately equal to Δy and is given by

$$dy = \frac{dy}{dx} \cdot \Delta x = 12x^{11} \cdot 1 = 12x^{11} \quad \therefore (401)_{12} = 20 + 0.025 = 20.025$$

Thus approximate value of $(401)_{12}$ is 20.025.

Question 3.

Find approximate value of $f(5.001)$ where $f(x) = x^3 - 7x^2 + 15$.

Solution:

$$\text{Here } f(x) = x^3 - 7x^2 + 15$$

$$f'(x) = 3x^2 - 14x$$

$$\text{Let } x = 5 \text{ and } x + \Delta x = 5.001$$

$$\therefore \Delta x = (x + \Delta x) - x = 5.001 - 5$$

$$= 0.001$$

$$f(5.001) = f(x + \Delta x)$$

$$\text{Now } \Delta y = f(x + \Delta x) - f(x)$$

$$f(x + \Delta x) = f(x) + \Delta y$$

$$= f(x) + f'(x) \cdot \Delta x \quad [\because \Delta y = f'(x) \cdot \Delta x]$$

$$= (x^3 - 7x^2 + 15) + (3x^2 - 14x)(0.001)$$

$$= (5^3 - 7 \times 5^2 + 15) + (3 \times 5^2 - 14 \times 5)[0.001] = [125 - 175 + 15] + [75 - 70]$$

$$[0.001] = -35 + 0.005 = -34.995$$

Thus approximate value of $f(5.001)$ is -34.995

Question 4.

If the radius of a sphere is measured as 7 m with an error of 0.02 m then find the approximate error in calculating its volume.

Solution:

Let r be the radius of the sphere and Δr be the error in measuring the radius.

$$\text{Then } r = 7 \text{ m and } \Delta r = 0.02 \text{ m}$$

Now volume of a sphere is given by

$$V = \frac{4}{3}\pi r^3 \Rightarrow dV = 4\pi r^2 dr \quad \therefore dV = \frac{dV}{dr} \cdot \Delta r = (4\pi r^2) \cdot \Delta r = 4\pi \times 7^2 \times 0.02 = 3.92\pi \text{ m}^3$$

Ex 8.2

Question 1.

Find differential dy for each of the following functions:

$$(i) y = (1 - 2x)^3 - 4x$$

$$(ii) y = (3 + \sin(2x))^{2/3}$$

$$(iii) y = e^{x^2 - 5x + 7} \cos(x^2 - 1)$$

Solution:

$$(i) y = (1 - 2x)^3 - 4x.$$

$$dy = [(3-4x)^3(1-2x)^2(-2) - (1-2x)^3(-4)(3-4x)^2]dx \text{ (i.e.,) } d$$

$$y = \{(1-2x)^2(3-4x)^2[-6(3-4x) + 4(1-2x)]\}dx \text{ (i.e.,) } dy = \{(1-2x)^2(3-4x)^2[16x-14]\}$$

$$dx = 2(8x-7)(1-2x)^2(3-4x)^2 dx \quad dy = 2(8x-7)(1-2x)^2(3-4x)^2 dx$$

(ii)

$$y = [3 + \sin(2x)]_{23} dy = \{23[3 + \sin(2x)]_{-13}[(\cos 2x)(2)]\} dx = \{43(3 + \sin 2x)_{-13}(\cos 2x)\}$$

$$dx = \left[\int \left[43 \cos 2x (3 + \sin 2x) \right]_{13} \right] dx$$

(iii)

$$y = e^{x^2-5x+7} \cos(x^2-1) dy = \{e^{x^2-5x+7}[-\sin(x^2-1)(2x)] + \cos(x^2-1)[e^{x^2-5x+7}(2x-5)]\}$$

$$dx dy = \{e^{x^2-5x+7}[-2x \sin(x^2-1) + (2x-5)\cos(x^2-1)]\}$$

$$dx dy = e^{x^2-5x+7}\{(2x-5)\cos(x^2-1) - 2x \sin(x^2-1)\} dx$$

Question 2 .

Find df for $f(x) = x^2 + 3x$ and evaluate it for

(i) $x=2$ and $dx=0.1$

(ii) $x=3$ and $dx=0.02$

Solution:

Solution:

$$y = f(x) = x^2 + 3x \quad dy = (2x+3)dx$$

$$(i) \quad dy \{ \text{when } x=2 \text{ and } dx=0.1 \} = [2(2)+3](0.1) = 7(0.1) = 0.7$$

$$(ii) \quad dy \{ \text{when } x=3 \text{ and } dx=0.02 \} = [2(3)+3](0.02) = 9(0.02) = 0.18$$

Question 3.

Find Δf and df for the function f for the indicated values of $x, \Delta x$ and compare

(i) $f(x) = x^3 - 2x^2; x=2, \Delta x = dx = 0.5$

(ii) $f(x) = x^2 + 2x + 3; x = -0.5, \Delta x = dx = 0.1$

Solution:

$$(i) \quad y = f(x) = x^3 - 2x^2$$

$$dy = (3x^2 - 4x)dx$$

$$dy \text{ (when } x=2 \text{ and } dx=0.5) = [3(2^2) - 4(2)](0.5)$$

$$= (12 - 8)(0.5) = 4(0.5) = 2$$

$$\text{(i.e.,) } df = 2$$

$$\text{Now } \Delta f = f(x + \Delta x) - f(x)$$

$$\text{Here } x=2 \text{ and } \Delta x=0.5$$

$$f(x) = x^3 - 2x^2$$

$$\text{So } f(x + \Delta x) = f(2 + 0.5) = f(2.5) = (2.5)^3 - 2(2.5)^2 = (2.5)^2[2.5 - 2] = 6.25(0.5) = 3.125$$

$$f(x) = f(2) = 2^3 - 2(2^2) = 8 - 8 = 0$$

$$\text{So } \Delta f = 3.125 - 0 = 3.125$$

$$(ii) \quad y = f(x) = x^2 + 2x + 3$$

$$dy = (2x + 2)dx$$

$$dy \text{ (when } x = -0.5 \text{ and } dx = 0.1)$$

$$= [2(-0.5) + 2](0.1)$$

$$= (-1 + 2)(0.1) = (1)(0.1) = 0.1$$

$$\text{(i.e.,) } df = 0.1$$

$$\text{Now } \Delta f = f(x + \Delta x) - f(x)$$

$$\text{Here } x = -0.5 \text{ and } \Delta x = 0.1$$

$$x^2 + 2x + 3$$

$$\begin{aligned}
 f(x+\Delta x) &= f(-0.5+0.1) = f(-0.4) \\
 &= (-0.4)^2 + 2(-0.4) + 3 \\
 &= 0.16 - 0.8 + 3 = 3.16 - 0.8 = 2.36 \\
 f(x) &= f(-0.5) = (-0.5)^2 + 2(-0.5) + 3 \\
 &= 0.25 - 1 + 3 = 3.25 - 1 = 2.25 \\
 \text{So } \Delta f &= f(x+\Delta x) - f(x) = 2.36 - 2.25 = 0.11
 \end{aligned}$$

Question 4.

Assuming $\log_{10} e = 0.4343$, find an approximate value of $\log_{10} 1003$.

Solution:

To find $\log 1003$

$$1003 = 1000 + 3 \text{ and}$$

$$\log 1000 = 3$$

Let $y = \log x$

$$dy = \frac{1}{x} dx$$

Here $x = 1000$ and $dx = 3 = 3 \log_{10} e$

$$\text{So } dy = \frac{1}{1000} [3 \times 0.4343] = 0.0013029$$

$$\text{Now } y + dy = \log 1000 + dy$$

$$= 3 + 0.0013029$$

$$= 3.0013029$$

Question 5.

The trunk of a tree has diameter 30 cm. During the following year, the circumference grew 6 cm.

(i) Approximately, how much did the tree's diameter grow?

(ii) What is the percentage increase in area of the tree's cross-section?

Solution:

(i) Given $r = 15$ cm and rate of change of perimeter = 6 cm

To find the rate of change of diameter

$$\text{Now perimeter } = p = 2\pi r$$

$$\text{So } dp = 2\pi dr$$

Here $dp = 6$ cm (given)

$$\Rightarrow 6 = 2\pi dr \Rightarrow 62\pi = dr \text{ (i.e.,) } dr = 3\pi$$

(i.e.,) rate of change in radius = $dr = 3\pi$

$$\Rightarrow \text{so rate of change in diameter} = d(2r) = 2dr = 2 \times 3\pi = 6\pi \text{ cm}$$

(ii) $A = \pi r^2$

$$A = \pi r^2 \text{ Now } dA = \pi(2r)dr = 2\pi r dr \text{ so } dA = 2\pi r dr \pi r^2 = 2dr r = 2(3\pi)r (=15) = 615\pi = 25\pi \text{ s}$$

$$\text{so \% error} = \frac{dA}{A} \times 100 = 25\pi \times 100 = 40\pi$$

Question 6.

An egg of a particular bird is very nearly spherical. If the radius to the inside of the shell is 5 mm and radius to the outside of the shell is 5.3 mm, find the volume of the shell approximately.

Solution:

$$V = \frac{4}{3}\pi r^3$$

Here $r = 5$ cm and $dr = 0.3$

$$\text{so } dv = 4\pi(25)310 = 30\pi \text{ mm}^3$$

Question 7.

Assume that the cross section of the artery of human is circular. A drug is given to a patient to

dilate his arteries. If the radius of an artery is increased from 2 mm to 2.1 mm, how much is cross-sectional area increased approximately?

Solution:

Area of circle $= A = \pi r^2$.

So $dA = \pi(2r)dr$ $dA = 2\pi r dr$ (i.e., $r = 2$ mm and $dr = 0.1$ mm) so $dA = 2\pi(2)(0.1) = 0.4\pi$

Question 8.

In a newly developed city, it is estimated that the voting population (in thousands) will increase according to $V(t) = 30 + 12t^2 - t^3$, $0 \leq t \leq 8$ where t is the time in years. Find the approximate change in voters for the time change from 4 to 4.16 year.

Solution:

$V = 30 + 12t^2 - t^3$ $0 \leq t \leq 8$

So $dV = (24t - 3t^2)dt$

Here $t = 4$ and $dt = 0.16$

So

$dV = [24(4) - 3(4^2)]0.16 = (96 - 48)0.16 = 48 \times 0.16 = 7.68$ (in thousand) $= 7680$

Question 9.

The relation between the number of words y a person learns in x hours is given by

$y = 52\sqrt{x}$, $0 \leq x \leq 9$. What is the approximate number of words learned when x changes from

(i) 1 to 1.1 hour?

(ii) 4 to 4.1 hour?

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Solution:

$y = 52\sqrt{x}$ $dy = 52(12\sqrt{x} dx) = 26\sqrt{x} dx$

(i) dy (when $x = 1$ and $dx = 0.1$ hr)

$= 26\sqrt{1}(0.1) = 2.6 \sim 3$ words

(ii)

dy (when $x = 4$ and $dx = 0.1$ hr) $= 26\sqrt{4}(0.1) = 262(0.1) = 13(0.1) = 1.3 \sim 1$ word

Question 10.

A circular plate expands uniformly under the influence of heat. If its radius increases from 10.5 cm to 10.75 cm, then find an approximate change in the area and the approximate percentage change in the area.

Solution:

Here radius is changing from 10.5 cm to 10.75 cm $\Rightarrow r = 10.5$ cm and $dr = 0.25$ cm

Now area $= A = \pi r^2 \Rightarrow dA = \pi(2r)dr$

(i) So dA

(when $r = 10.5$ cm and $dr = 0.25$ cm)

$= \pi(2 \times 10.5)(0.25)$

$= 5.25\pi$

(ii) Percentage error in Area $= \frac{dA}{A} \times 100$

(i.e.,) Here $dA = 2\pi r dr$ $A = \pi r^2$

So $\frac{dA}{A} = \frac{2\pi r dr}{\pi r^2} = \frac{2dr}{r}$

$\frac{dA}{A} \times 100$ (when $r = 10.5$ and $dr = 0.25$ cm)

$$=2(0.25)^{10.5} \times 100$$

$$=4.76\%$$

Question 11.

A coat of paint of thickness 0.2 cm is applied to the faces of a cube whose edge is 10 cm. Use the differentials to find approximately how many cubic centimeters of paint is used to paint this cube. Also calculate the exact amount of paint used to paint this cube.

Solution:

$$(i) v = a^3$$

$$\text{so } dv = a^2 da$$

$$dv \text{ (when) } a = 10 \text{ cm and } da = 0.20 \text{ cm}$$

$$= 3(10^2)(0.2)$$

$$300 \times 0.2 = 60 \text{ cm}^3$$

$$\text{Actual paint used} = v \text{ at } x + \Delta x = 10.2 \text{ and } x = 10 \text{ cm}$$

$$= a^3 \text{ at } x + \Delta x = 10.2 \text{ and } x = 10$$

$$= (10.2)^3 - (10)^3 = 61.2 \text{ cm}^3$$

Ex 8.3

Question 1.

Evaluate $\lim_{(x,y) \rightarrow (1,2)} g(x,y)$, if the limit exists, where $g(x,y) = 3x^2 - xyx^2 + y^2 + 3$.

Solution:

$$\lim_{(x,y) \rightarrow (1,2)} g(x,y) = \lim_{(x,y) \rightarrow (1,2)} (3x^2 - xyx^2 + y^2 + 3) = \lim_{(x,y) \rightarrow (1,2)} 3x^2 - xy \lim_{(x,y) \rightarrow (1,2)} x^2 + y^2 + 3$$

$$= 3(1)^2 - (1)(2)(1)^2 + (2)^2 + 3 = 3 - 2 + 4 + 3 = 8$$

If $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$ then the limit exists.

Question 2.

Evaluate $\lim_{(x,y) \rightarrow (0,0)} \cos(x^3 + y^2x + y + 2)$. If the limit exists.

Solution:

$$\lim_{(x,y) \rightarrow (0,0)} \cos(x^3 + y^2x + y + 2)$$

$$\text{Let us take } y = 0 \text{ for } f(y), \text{ then } \lim_{y \rightarrow 0} \cos(y^2y + 2) = 1 \neq \infty$$

$\therefore$ limit exists

$$\text{Now, } \lim_{(x,y) \rightarrow (0,0)} \cos(x^3 + y^2x + y + 2) = \cos(0^3 + 0^2 \cdot 0 + 0 + 2) = \cos(2) = \cos(0) = 1$$

Question 3.

Let $f(x,y) = y^2 - xy\sqrt{x} - \sqrt{y}$ for $(x,y) \neq (0,0)$. Show that $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$.

Solution:

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} (y^2 - xy\sqrt{x} - \sqrt{y}) = \lim_{(x,y) \rightarrow (0,0)} y^2 - xy\sqrt{x} - \sqrt{y} \times \sqrt{x} + \sqrt{x} + \sqrt{y}$$

$$= \lim_{(x,y) \rightarrow (0,0)} y^2 - y(x-y)(\sqrt{x} + \sqrt{y})(x-y) = \lim_{(x,y) \rightarrow (0,0)} y(\sqrt{x} + \sqrt{y}) = 0$$

Hence proved

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Question 4.

Evaluate $\lim_{(x,y) \rightarrow (0,0)} \cos(e^x \sin y)$, if the limit exists.

Solution:

$$(x,y) \rightarrow (0,0) \cos(e^x \sin y)$$

Using two different parts to approach the point along $x=0$ and $y=0$

$$\lim_{(x,y) \rightarrow (0,0)} \cos(e^x \sin y) = \cos(\lim_{(x,y) \rightarrow (0,0)} e^x \sin y) = \cos(1) (\because \lim_{y \rightarrow 0} \sin y = 0)$$

Question 5.

Let $g(x,y) = x^2y^4 + y^2$ for $(x,y) \neq (0,0)$ and $f(0,0) = 0$.

(i) Show that $\lim_{(x,y) \rightarrow (0,0)} g(x,y) = 0$ along every line $y = mx, m \in \mathbb{R}$.

(ii) Show that $\lim_{(x,y) \rightarrow (0,0)} g(x,y) = k_1 + k_2$ along every parabola $y = kx^2, k \in \mathbb{R} \setminus \{0\}$.

Solution:

Given $g(x,y) = x^2 y x^4 + y^2$ for $(x,y) \neq (0,0)$ and $f(0,0) = 0$

(i) Along the line $y = mx, m \in \mathbb{R}$

$$\lim_{(x,y) \rightarrow (0,0)} g(x,y) = \lim_{(x,mx) \rightarrow (0,0)} (x^2(mx)x^4 + (mx)^2) = \lim_{(x,mx) \rightarrow (0,0)}$$

$$(mx^3x^4 + m^2x^2) = \lim_{(x,mx) \rightarrow (0,0)} (mxx^2 + m^2) = 0m^2 = 0$$

Hence Proved.

(ii) $g(x,y) = x^2 y x^4 + y^2$ along every parabola $y = kx^2, k \in \mathbb{R} \setminus \{0\}$

$$\lim_{(x,y) \rightarrow (0,0)} g(x,y) = \lim_{(x,kx^2) \rightarrow (0,0)} x^2(kx^2)x^4 +$$

$$(kx^2)^2 = \lim_{(x,kx^2) \rightarrow (0,0)} kx^4x^4(1+k^2) = \lim_{(x,kx^2) \rightarrow (0,0)} (k^1+k^2) = k^1+k^2$$

Hence Proved.

Question 6.

Show that $f(x,y) = x^2 - y^2 y^2 + 1$ is continuous at every $(x,y) \in \mathbb{R}^2$.

Solution:

$$f(x,y) = x^2 - y^2 y^2 + 1$$

Let $(a,b) \in \mathbb{R}^2$ be an arbitrary point.

Now $f(a,b) = a^2 - b^2 b^2 + 1$ is defined

$$\lim_{(x,y) \rightarrow (a,b)} x^2 - y^2 y^2 + 1 = \lim_{(x,y) \rightarrow (a,b)} [x^2 - y^2] \lim_{(x,y) \rightarrow (a,b)} [y^2 + 1] = a^2 - b^2 b^2 + 1 = L \text{ limit}$$

$$\text{exists } \lim_{(x,y) \rightarrow (a,b)} f(x,y) = L = f(a,b)$$

Here, f satisfies all the three conditions of continuity at (a, b) . Hence, f is continuous at every point of $\mathbb{R}^2$ as $(a,b) \in \mathbb{R}^2$.

Question 7.

Let $g(x,y) = e^{y \sin x}$, for $x \neq 0$ and $g(0,0) = 1$. Show that g is continuous at $(0,0)$.

Solution:

$$g(x,y) = e^{y \sin x}, x \neq 0 \text{ and } g(0,0) = 1$$

By definition $g(0,0) = 1$ (given)

(i.e.,)

$$\lim_{(x,y) \rightarrow (0,0)} g(x,y) = \lim_{(x,y) \rightarrow (0,0)} e^{y \sin x} = [\lim_{(x,y) \rightarrow (0,0)} e^y][x \lim_{(x,y) \rightarrow (0,0)} \sin x] = (1)(1) = 1$$

Now, limit along the line $y = mx$, passing through $(0,0)$

$$\lim_{(x,mx) \rightarrow (0,0)} e^{mx \sin x} = 1 \text{ (defined)}$$

Hence, ' g ' is continuous at $(0,0)$

Ex 8.4

Question 1.

Find the partial derivatives of the following functions at the indicated points.

(i) $f(x,y) = 3x^2 - 2xy + y^2 + 5x + 2, (2, -5)$

(ii) $g(x,y) = 3x^2 + y^2 + 5x + 2, (1, -2)$

(iii) $h(x,y,z) = x \sin(xy) + z^2 x, (2, \pi/4, 1)$

(iv) $G(x,y) = e^{x+3y} \log(x^2 + y^2), (-1, 1)$

Solution:

(i)

$$f(x,y) = 3x^2 - 2xy + y^2 + 5x + 2 \quad \partial f / \partial x = 6x - 2y + 5, \quad \partial f / \partial y = -2x + 2y \quad \text{At } (2, -5) \Rightarrow \partial f / \partial x = 6(2) - 2(-5) + 5 = 27$$

$$\partial f / \partial y = -2(2) + 2(-5) = -14$$

(ii)

$$g(x,y)=3x^2+y^2+5x+2$$

$$\partial g/\partial x=6x+5, \partial g/\partial y=2y$$

$$\text{At } (1, -2) \Rightarrow \partial g/\partial x=6(1)+5=11, \partial g/\partial y=2(-2)=-4$$

(iii)

$$h(x,y,z)=x\sin(xy)+z^2x\partial h/\partial x=x[y\cos(xy)]+\sin(xy)+z^2\partial h/\partial y=x^2\cos(xy)$$

$$\text{At } \partial h/\partial z=2zx\partial h/\partial x=2[\pi^4, 1)\cos(2\pi^4)]$$

$$+\sin(2\pi^4)+(1)^2=\pi^2\cos(\pi^2)+\sin(\pi^2)+1=\pi^2(0)+1+1=2, \partial h/\partial y=(2)^2\cos(2\pi^4)=4\cos(\pi^2)=4(0)=0, \partial h/\partial z=2(1)(2)=4$$

(iv)

$$G(x,y)=e^{x+3y}\log(x^2+y^2), \partial G/\partial x=e^{x+3y}[2xx^2+y^2]+\log(x^2+y^2)$$

$$[e^{x+3y}]=e^{x+3y}[2xx^2+y^2+\log(x^2+y^2)], \partial G/\partial y=e^{x+3y}[2yx^2+y^2]+\log(x^2+y^2)$$

$$[3e^{x+3y}], \partial G/\partial y=e^{x+3y}[2yx^2+y^2+3\log(x^2+y^2)], \partial G/\partial x=e^{-1+3}[-2+1+\log(1+1)]=e^2[-1+\log 2]$$
$$]=e^2[\log 2-1], \partial G/\partial y=e^{-1+3}[2+1+3\log(1+1)]=e^2[1+3\log 2]=e^2[1+\log 8]$$

Question 2.

For each of the following functions find the f_x, f_y and show that $f_{xy}=f_{yx}$

(i) $f(x,y)=3xy+\sin x$

(ii) $f(x,y)=\tan^{-1}(xy)$

(iii) $f(x,y)=\cos(x^2-3xy)$

Solution:

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(i) $f(x,y)=3xy+\sin x, f_x=(y+\sin x)[3]-3x[0+\cos x]$

$$(y+\sin x)^2=3y+3\sin x-3x\cos x, (y+\sin x)^2 f_y=(y+\sin x)[0]-3x[1+0]$$

$$(y+\sin x)^2=-3x(y+\sin x)^2, \partial^2 f/\partial x \partial y=\partial \partial x[-3x(y+\sin x)^2]=(y+\sin x)^2[-3]-$$

$$(-3x)^2(y+\sin x)[0+\cos x](y+\sin x)^4=-3(y+\sin x)^2+6x\cos x(y+\sin x)$$

$$(y+\sin x)^4, \partial^2 f/\partial y \partial x=\partial \partial y[3y+3\sin x-3x\cos x]$$

$$(y+\sin x)^2=(y+\sin x)^2[3]-(3y+3\sin x-3x\cos x)^2(y+\sin x)(0+\cos x)$$

$$(y+\sin x)^4=-3(y+\sin x)^2+6x\cos x(y+\sin x)(y+\sin x)^4 \dots (2)$$

$$\text{from (1) \& (2)} \Rightarrow \partial^2 f/\partial x \partial y=\partial^2 f/\partial y \partial x$$

(ii) $f(x,y)=\tan^{-1}(xy), \partial f/\partial x=1/(1+x^2y^2)(1y)=yx^2+y^2, \partial^2 f/\partial y=\partial^2 f/\partial x \partial y=1/(1+x^2y^2)(-xy^2)=-xx^2+y^2, \partial^2 f/\partial x \partial y=$

$$\partial \partial x[\partial f/\partial y]=\partial \partial x[-xx^2+y^2]=(x^2+y^2)[-1]-(-x)[2x]$$

$$(x^2+y^2)^2=x^2-y^2, (x^2+y^2)^2 \partial^2 f/\partial y \partial x=\partial \partial y[\partial f/\partial x]=\partial \partial y[yx^2+y^2]=(x^2+y^2)[1]-y[2y]$$

$$(x^2+y^2)^2=x^2-y^2, (x^2+y^2)^2 \text{ from (1) \& (2)} \Rightarrow \partial^2 f/\partial x \partial y=\partial^2 f/\partial y \partial x$$

(iii) $f(x,y)=\cos(x^2-3xy), \partial f/\partial x=-\sin(x^2-3xy)$

$$[2x-3y]=(3y-2x)\sin(x^2-3xy), \partial f/\partial y=-\sin(x^2-3xy)$$

$$[0-3x]=3x\sin(x^2-3xy), \partial^2 f/\partial x \partial y=\partial \partial x[\partial f/\partial y]=\partial \partial x[3x\sin(x^2-3xy)]=3x[\cos(x^2-3xy) \cdot (2x-3y)+\sin(x^2-3xy)[3]]=3x(2x-3y)\cos(x^2-3xy)+3\sin(x^2-3xy) \dots$$

$$(1)=\partial \partial y[\partial f/\partial x]=\partial \partial y[(3y-2x)\sin(x^2-3xy)]=(3y-2x)[\cos(x^2-3xy) \cdot (-3x)]$$

$$+\sin(x^2-3xy)[3]=3x(2x-3y)\cos(x^2-3xy)+3\sin(x^2-3xy) \dots (2) \text{ from (1) \&}$$

$$(2) \Rightarrow \partial^2 f/\partial x \partial y=\partial^2 f/\partial y \partial x$$

Question 3.

If $U(x,y,z)=x^2+y^2xy+3z^2y$, find $\partial U/\partial x, \partial U/\partial y$ and $\partial U/\partial z$

Solution:

$$U(x,y,z)=x^2+y^2xy+3z^2y, \partial U/\partial x=(xy)(2x)-(x^2+y^2)(y)$$

$$(xy)^2 + 0 = 2x_2y - x_2y - y_3(xy)^2 = x_2y - y_3(xy)^2 = y(x_2 - y_2)x_2y^2 = x_2 - y_2x_2y^2 \partial U \partial y = (xy)(2y) - (x_2 + y_2)(x)$$

$$(xy)^2 + 3z^2 = 2xy^2 - x_3 - y_2x(xy)^2 + 3z^2 = xy^2 - x_3(xy)^2 + 3z^2 = x(y_2 - x_2)x_2y^2 = y_2 - x_2y_2x + 3z^2 \partial U \partial z = 0 + 6zy = 6zy$$

Question 4.

If $U(x,y,z) = \log(x^2 + y^3 + z^3)$, find $\partial U \partial x + \partial U \partial y$ and $\partial U \partial z$.

Solution:

$$U(x,y,z) = \log(x^2 + y^3 + z^3) \partial U \partial x = 3x_2x_3 + y_3 + z_3 \partial U \partial y = 3y_2x_3 + y_3 + z_3 \partial U \partial z = 3z_2x_3 + y_3 + z_3 \partial U \partial x + \partial U \partial y + \partial U \partial z = 3(x_2 + y_2 + z_2)(x_3 + y_3 + z_3)$$

Question 5.

For each of the following functions find the g_{xy}, g_{xx}, g_{yy} and g_{yx} .

(i) $g(x,y) = x_y + 3x_2y$

(ii) $g(x,y) = \log(5x + 3y)$

(iii) $g(x,y) = x^2 + 3xy - 7y + \cos(5x)$

Solution:

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(i)

$$\partial g \partial x = g_x = e_y + 6xy \partial g \partial y = g_y = x e_y + 3x_2 g_{xx} = \partial^2 g \partial x^2 = \partial \partial x [\partial g \partial x] = \partial \partial x [e_y + 6xy] = 0 + 6y = 6y g_{yy} = \partial^2 g \partial y^2 = \partial \partial y [\partial g \partial y] = \partial \partial y [x e_y + 3x_2] = x e_y g_{xy} = \partial^2 g \partial x \partial y = \partial \partial x [\partial g \partial y] = \partial \partial x [x e_y + 3x_2] = e_y + 6x g_{yx} = \partial^2 g \partial y \partial x = \partial \partial y [\partial g \partial x] = \partial \partial y [e_y + 6xy] = e_y + 6x$$

(ii)

$$g_x = \partial g \partial x = 15x + 3y(5) = 55x + 3y g_y = \partial g \partial y = 15x + 3y(3) = 35x + 3y$$

$$g_{xx} = \partial^2 g \partial x^2 = \partial \partial x [\partial g \partial x] = \partial \partial x [55x + 3y] = (5x + 3y)(0) - 5(5)$$

$$(5x + 3y)^2 = -25(5x + 3y)^2 g_{yy} = \partial^2 g \partial y^2 = \partial \partial y [\partial g \partial y] = \partial \partial y [35x + 3y] = (5x + 3y)(0) - 3(3)$$

$$(5x + 3y)^2 = -9(5x + 3y)^2 g_{xy} = \partial^2 g \partial x \partial y = \partial \partial x [\partial g \partial y] = \partial \partial x [35x + 3y] = -3(5x + 3y)^2(5) = -15(5x + 3y)^2$$

(iii)

$$g(x,y) = x^2 + 3xy - 7y + \cos(5x) g_x = \partial g \partial x = 2x + 3y - 5\sin(5x) g_y = \partial g \partial y = 3x - 7 g_{xx} = 2 - 25\cos(5x) g_{yy} = 0 g_{xy} = \partial \partial x (\partial g \partial y) = \partial \partial x (3x - 7) = 3 g_{yx} = \partial \partial y (\partial g \partial x) = \partial \partial y (2x + 3y - 5\sin(5x)) = 3$$

Question 6.

Let $w(x,y,z) = 1\sqrt{x^2 + y^2 + z^2}$ ($x,y,z \neq (0,0,0)$). Show that $\partial^2 w \partial x^2 + \partial^2 w \partial y^2 + \partial^2 w \partial z^2 = 0$

Solution:

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$$w(x,y,z) = 1\sqrt{x^2 + y^2 + z^2} \partial w \partial x = -12(2x)$$

$$(x^2 + y^2 + z^2)^{3/2} = -x(x^2 + y^2 + z^2)^{3/2} \partial w \partial y = -y(x^2 + y^2 + z^2)^{3/2} \partial^2 w \partial x^2 = (x^2 + y^2 + z^2)^{3/2}(-1) + x^2(x^2 + y^2 + z^2)^{1/2}(2x)[(x^2 + y^2 + z^2)^{3/2}]^2 = (x^2 + y^2 + z^2)^{1/2}[-x^2 - y^2 - z^2 + 3x^2]$$

$$(x^2 + y^2 + z^2)^3 = 2x^2 - y^2 - z^2(x^2 + y^2 + z^2)^{5/2} \partial^2 w \partial y^2 = -x^2 + 2y^2 - z^2(x^2 + y^2 + z^2)^{5/2} \partial^2 w \partial z^2 = -x^2 - y^2 + 2z^2(x^2 + y^2 + z^2)^{5/2}$$

$$\partial^2 w \partial x^2 + \partial^2 w \partial y^2 + \partial^2 w \partial z^2 = 0(x^2 + y^2 + z^2)^{5/2} \partial^2 w \partial x^2 + \partial^2 w \partial y^2 + \partial^2 w \partial z^2 = 0$$

Question 7.

If $V(x,y) = e_x(x\cos y - y\sin y)$, then prove that $\partial^2 V \partial x^2 + \partial^2 V \partial y^2 = 0$

Solution:

$$V(x,y) = e_x[x\cos y - y\sin y] V = x e_x \cos y - e_x y \sin y \partial V \partial x = \cos y [e_x + x e_x] - e_x y \sin y \partial^2 V \partial x^2 = \cos y [e_x + e_x + x e_x] - e_x y \sin y \partial^2 V \partial x^2 = e_x \cos y + e_x \cos y + x e_x \cos y - e_x y \sin y \partial V \partial y = x e_x [-\sin y] - e_x [y \cos y + \sin y] \partial^2 V \partial x^2 = 0$$

$$\partial y_2 = -x e_x \cos y - e_x [-y \sin y + \cos y + \cos y] \partial_2 V \partial y_2 = -x e_x \cos y + e_x y \sin y - e_x \cos y - e_x \cos y(1) + (2) \Rightarrow \partial_2 V \partial x_2 + \partial_2 V \partial y_2 = 0$$

Question 8.

If $w(x,y) = xy + \sin(xy)$, then prove that $\partial_2 w \partial y \partial x = \partial_2 w \partial x \partial y$

Solution:

$$W(x,y) = xy + \sin(xy) \text{ LHS} = \partial_2 w \partial y \partial x = \partial \partial y [\partial w \partial x] = \partial \partial y [y + y \cos(xy)] = 1 + y[-x \sin(xy)] = 1 - x y \sin(xy) \text{ RHS} = \partial_2 w \partial x \partial y = \partial \partial x [\partial w \partial y] \text{ from (1) \& (2) } \Rightarrow \partial_2 w \partial y \partial x = \partial_2 w \partial x \partial y$$

Question 9.

If $v(x,y,z) = x^3 + y^3 + z^3 + 3xyz$, show that $\partial_2 v \partial y \partial z = \partial_2 v \partial z \partial y$

Solution:

$$v(x,y,z) = x^3 + y^3 + z^3 + 3xyz \partial_2 v \partial y \partial z = \partial \partial y [\partial v \partial z] = \partial \partial y [3z^2 + 3xy] = 3x \partial_2 v \partial z \partial y = \partial \partial z [\partial v \partial y] = \partial \partial z [3y^2 + 3xz] = 3x$$

$$\text{from (1) \& (2) } \Rightarrow \partial_2 v \partial y \partial z = \partial_2 v \partial z \partial y$$

Question 10.

A firm produces two types of calculators each week, x number of type A and y number of type B. The weekly revenue and cost functions (in rupees) are

$$R(x,y) = 80x + 90y + 0.04xy - 0.05x^2 - 0.05y^2 \text{ and } C(x,y) = 8x + 6y + 2000 \text{ respectively.}$$

(i) Find the profit function $P(x,y)$,

(ii) Find $\partial P \partial x(1200, 1800)$ and $\partial P \partial y(1200, 1800)$ and interpret these results.

Solution:

(i) Profit =

$$\text{Revenue} - \text{Cost} = (80x + 90y + 0.04xy - 0.05x^2 - 0.05y^2) - (8x + 6y + 2000) = 80x + 90y + 0.04xy - 0.05x^2 - 0.05y^2 - 8x - 6y - 2000 \\ P(x,y) = 72x + 84y + 0.04xy - 0.05x^2 - 0.05y^2 - 2000$$

(ii)

$$\partial P \partial x = 72 + 0.04y - 0.1x \partial P \partial y = 84 + 0.04x - 0.1y \text{ At } (1200, 1800) \partial P \partial x = 72 + 0.04(1800) - 0.1(1200) \text{ At } (1200, 1800) = 72 + 72 - 120 = 24 \\ \partial P \partial x = 72 + 0.04(1800) - 0.1(1200) \partial P \partial y = 72 + 72 - 120 = 24 = 84 + 0.04(1200) - 0.1(1800) 84 + 48 - 180 = -48$$

Ex 8.5

Question 1.

If $w(x,y) = x^3 - 3xy + 2y^2, x,y \in \mathbb{R}$, find the linear approximation for w at $(1, -1)$.

Solution:

$$w(x,y) = x^3 - 3xy + 2y^2; \text{ at } (1, -1)$$

Linear approximation is given by

$$L(x,y) = w(x_0, y_0) + (\partial w \partial x)_{(x_0, y_0)}(x - x_0) + (\partial w \partial y)_{(x_0, y_0)}(y - y_0)$$

$$\text{Here } (x_0, y_0) = (1, -1) \partial W \partial x = 3x^2 - 3y \partial W \partial y = -3x + 4y \text{ At } (1, -1) w = (1)^3 - 3(1)(-1) + 2(-1)^2 = 6$$

$$\partial W \partial x = 3(1)^2 - 3(-1) = 6 \partial W \partial y = -3(1) + 4(-1) = -7 \text{ (1) } \Rightarrow L(x,y) = 6 + 6(x - 1) - 7(y + 1) = 6x - 6 - 7y - 7 = 6x - 7y - 7$$

Question 2 .

Let $z(x,y) = x^2y + 3xy^4, x,y \in \mathbb{R}$, Find the linear approximation for z at $(2, -1)$.

Solution:

$$z(x,y) = x^2y + 3xy^4 \text{ at } (2, -1) \text{ Here } (x_0, y_0) = (2, -1) \partial z \partial x = 2xy + 3y^4 \partial z \partial y = x^2 + 12xy^3 \text{ At } (2, -1) = (2)^2(-1) + 3(2)(-1)^4 = -4 + 6 = 2 \\ \partial z \partial x = 2(2)(-1) + 3(-1)^4 = -4 + 3 = -1 \partial z \partial y = (2)^2 + 12(2)(-1)^3 = 4 - 24 = -20$$

Linear approximation is given by

$$L(x,y)=z(x_0,y_0)+(\partial z/\partial x)_{(x_0,y_0)}(x-x_0)+(\partial z/\partial y)_{(x_0,y_0)}(y-y_0)=2+(-1)(x-2)-20(y+1)=2-x+2-20y-20=-x-20y-16=-(x+20y+16)$$

Question 3.

If $v(x,y)=x^2-xy+14y^2+7$, $x,y \in \mathbb{R}$, find the differential dv .

Solution:

$$v(x,y)=x^2-xy+14y^2+7, v_x=2x-y, v_y=-x+14(2y)=-x+y^2$$

The differential is $dv=(2x-y)dx+(-x+y^2)dy$

Question 4.

Let $W(x,y,z)=x^2-xy+3\sin z$, $x,y,z \in \mathbb{R}$. Find the linear approximation at $(2,-1,0)$.

Solution:

$$w(x,y,z)=x^2-xy+3\sin z \text{ Here } (x_0,y_0,z_0)=(2,-1,0) \partial W/\partial x=2x-y \partial W/\partial y=-x \partial W/\partial z=3\cos z$$

$$\text{At } (2,-1,0) w=(2)^2-2(-1)+3\sin(0)=4+2+3(0)=6 \partial W/\partial x=2(2)-$$

$$(-1)=4+1=5 \partial W/\partial y=-(-2)=2 \partial W/\partial z=3\cos(0)=3(1)=3$$

Linear approximation is given by

$$L(x,y,z)=w(x_0,y_0,z_0)+(\partial W/\partial x)_{(x_0,y_0,z_0)}(x-x_0)+(\partial W/\partial y)_{(x_0,y_0,z_0)}(y-y_0)+(\partial W/\partial z)_{(x_0,y_0,z_0)}(z-z_0)$$

$$L(x,y,z)=6+5(x-2)+2(y+1)+3(z-0)=6+5x-10-2y-2+3z=5x-2y+3z-6$$

Question 5.

Let $V(x,y,z)=xy+yz+zx$, $x,y,z \in \mathbb{R}$. Find the differential dV .

Solution:

$$V(x,y,z)=xy+yz+zx, V_x=y+z$$

$$V_y=x+z, V_z=y+x$$

The differential is $dV=(y+z)dx+(x+z)dy+(y+x)dz$

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Ex 8.6

Question 1.

If $u(x,y)=x^2y+3xy^4$, $x=e^t$ and $y=\sin t$, find du/dt and evaluate it at $t=0$.

Solution:

$$u(x,y)=x^2y+3xy^4; x=e^t; y=\sin t \quad du/dt=\partial u/\partial x dx/dt+\partial u/\partial y dy/dt=(2xy+3y^4)(e^t)+(x^2+12xy^3)$$

$$(\cos t)=(2e^t \sin t+3\sin^4 t)e^t+[e^{2t}+12e^t \sin^3 t]\cos t=e^t[2e^t \sin t+3\sin^4 t+e^t(\cos t)$$

$$+12\sin^3 t \cos t] \text{ at } t=0 \quad du/dt=e^0[2e^0 \sin(0)+3\sin^4(0)+e^0 \cos(0)+12\sin^3(0)\cos(0)]=1[0+0+1+0]=1e^0=1 \cos 0=1 \sin 0=0$$

Question 2.

If $u(x,y,z)=xy^2z^3$, $x=\sin t$, $y=\cos t$, $z=1+e^{2t}$, find du/dt

Solution:

$$u=xy^2z^3; x=\sin t; y=\cos t; z=1+e^{2t} \quad du/dt=\partial u/\partial x dx/dt+\partial u/\partial y dy/dt+\partial u/\partial z dz/dt=(y^2z^3)(\cos t)$$

$$+(2xy^2z^3)(-\sin t)+(3xy^2z^2)(0+2e^{2t})=yz^2[yz \cos t-2xz \sin t+6xye^{2t}] \text{ (Replace$$

$$x=\sin t; y=\cos t \text{ and } z=1+e^{2t}) = \cos t(1+e^{2t})^2[\cos^2 t(1+e^{2t})-2\sin^2 t(1+e^{2t})$$

$$+6\sin t \cos t e^{2t}] \quad du/dt=(1+e^{2t})^2[\cos^3 t(1+e^{2t})-2\sin^2 t \cos t(1+e^{2t})+6e^{2t} \sin t \cos^2 t]$$

Question 3.

If $w(x,y,z)=x^2+y^2+z^2$, $x=e^t$, $y=e^t \sin t$, and $z=e^t \cos t$, find dw/dt

Solution:

$$w(x,y,z)=x^2+y^2+z^2; x=e^t; y=e^t \sin t; z=e^t \cos t \quad dw/dt=\partial w/\partial x dx/dt+\partial w/\partial y dy/dt+\partial w/\partial z dz/dt=(2$$

$$x)(e^t)+(2y)(e^t \cos t+e^t \sin t)+(2z)(e^t \cos t-e^t \sin t)=2e^t[x+y \cos t+y \sin t+z \cos t-z \sin t] \text{ (Re$$

place x, y, z - values) $= 2e^t[e^t + e^t \sin t \cos t + e^t \sin^2 t + e^t \cos^2 t - e^t \cos t \sin t] = 2e^t \cdot e^t [1 + \sin^2 t + \cos^2 t] = 2e^{2t} [2] = 4e^{2t}$

Question 4.

Let $U(x, y, z) = xyz$, $x = e^{-t}$, $y = e^{-t} \cos t$, $z = \sin t$, $t \in \mathbb{R}$. Find dU/dt .

Solution:

$U(x, y, z) = xyz$; $x = e^{-t}$; $y = e^{-t} \cos t$; $z = \sin t$
 $dU/dt = \partial U / \partial x \cdot dx/dt + \partial U / \partial y \cdot dy/dt + \partial U / \partial z \cdot dz/dt = (yz)(-e^{-t}) + (xz)(-e^{-t} \cos t - e^{-t} \sin t) + (xy)(\cos t) = e^{-t}[-yz - xz \cos t - xz \sin t + xycos t]$
 (Replace x, y, z value) $= e^{-t}[-e^{-t} \cos t \sin t - e^{-t} \sin t \cos t - e^{-t} \sin^2 t + e^{-t} \cdot e^{-t} \cdot \cos^2 t] = -e^{-t} \cdot e^{-t} [\cos t \sin t + \sin t \cos t + \sin^2 t - \cos^2 t] = -e^{-2t} [2 \cos t \sin t - (\cos^2 t - \sin^2 t)] = -e^{-2t} [\sin 2t - \cos 2t]$

Question 5.

If $w(x, y) = 6x^3 - 3xy + 2y^2$, $x = e^s$, $y = \cos s$, $s \in \mathbb{R}$, find dw/ds , and evaluate at $s = 0$.

Solution:

$w(x, y) = 6x^3 - 3xy + 2y^2$; $x = e^s$; $y = \cos s$
 $dw/ds = \partial w / \partial x \cdot dx/ds + \partial w / \partial y \cdot dy/ds = (18x^2 - 3y)(e^s) + (-3x + 4y)(-\sin s)$

(Replace x, y value)

$= 18e^{3s} - 3\cos s e^s + 3e^s \sin s - 4\sin s \cos s$ At $s = 0$ $dw/ds = 18 - 3 + 0 - 0 = 15$

Question 6.

If $z(x, y) = x \tan^{-1}(xy)$, $x = t^2$, $y = se^t$, $s, t \in \mathbb{R}$, Find $\partial z / \partial t$ and $\partial z / \partial s$ at $s = t = 1$

Solution:

$z(x, y) = x \tan^{-1}(xy)$; $x = t^2$; $y = se^t$

(i) $\partial z / \partial s = \partial z / \partial x \cdot \partial x / \partial s + \partial z / \partial y \cdot \partial y / \partial s = [xy^1 + (xy)^2 + \tan^{-1}(xy)](0) + x^2 1 + (xy)^2 (e^t) = x^2 e^t 1 + (xy)^2$
 (Replace x, y values) $\partial z / \partial s = e^{t^4} 1 + (t^2 se^t)^2$ At $s = 1$; $t = 1$ $x = 1$; $y = e$ $\partial z / \partial s = e^1 + e^2$

(ii)

$\partial z / \partial t = \partial z / \partial x \cdot \partial x / \partial t + \partial z / \partial y \cdot \partial y / \partial t = [xy^1 + x^2 y^2 + \tan^{-1}(xy)](2t) + [x^2 1 + x^2 y^2](se^t)$

At $s = 1$; $t = 1$

- $x = 1$; $y = e$

$\partial z / \partial t = 2[e^1 + e^2 + \tan^{-1}(e)] + [1 + e^2]e = 2e^1 + e^2 + 2 \tan^{-1} e + e^1 + e^2 = 3e^1 + e^2 + 2 \tan^{-1} e$

Question 7.

Let $U(x, y) = e^x \sin y$, where $x = st^2$, $y = s^2 t$, $s, t \in \mathbb{R}$. Find them at $s = t = 1$.

Solution:

(i)

$U(x, y) = e^x \sin y$; $x = st^2$; $y = s^2 t$

$\partial U / \partial s = \partial U / \partial x \cdot \partial x / \partial s + \partial U / \partial y \cdot \partial y / \partial s = (e^x \sin y)(t^2) + (e^x \cos y)(2st) = te^x [t \sin y + 2s \cos y]$

(Replace x, y values) $\partial U / \partial s = te^{st^2} [t \sin(s^2 t) + 2s \cos(s^2 t)]$ At $s = t = 1$ $\partial U / \partial s = e [\sin(1) + 2 \cos(1)]$

(ii)

$\partial U / \partial t = \partial U / \partial x \cdot \partial x / \partial t + \partial U / \partial y \cdot \partial y / \partial t$

$= (e^x \sin y)(2st) + (e^x \cos y)(s^2)$

$= s^x [2t \sin y + s \cos y]$

(Replace x, y values)

$\partial U / \partial t = se^{st^2} [2t \sin(s^2 t) + s \cos(s^2 t)]$

At $s = t = 1$

$\partial U / \partial t = e [2 \sin(1) + \cos(1)]$

Question 8.

Let $z(x, y) = x^3 - 3x^2 y^3$, where $x = st$, $y = s - t$, $s, t \in \mathbb{R}$. Find $\partial z / \partial s$ and $\partial z / \partial t$

Solution:

(i)

$$z(x,y)=x^3-3x^2y^3; x=se^t; y=se^{-t} \quad \partial U \partial t = \partial z \partial x \partial x \partial s + \partial z \partial y \partial y \partial s = (3x^2-6xy^3)(e^t) + (-9x^2y^2)(e^{-t}) = (e^t)[3x^2-6xy^3] - (e^{-t})[9x^2y^2]$$

(Replace x,y values)

$$\partial z \partial s = e^t[3(se^t)^2 - 6(se^t)(se^{-t})^3] - e^{-t}[9(se^t)^2(se^{-t})^2] = e^t[3s^2e^{2t} - 6s^4e^{t-3t}] - e^{-t}[9s^4] = e^t(3s^2)[e^{2t} - 2s^2e^{-2t} - 3s^2e^{-t}] = 3s^2e^t[e^{2t} - 2s^2e^{-2t} - 3s^2e^{-t}]$$

(ii)

$$\partial z \partial t = \partial z \partial x \partial x \partial t + \partial z \partial y \partial y \partial t = (3x^2-6xy^3)(se^t) + (-9x^2y^2)(-se^{-t})$$

(Replace x,y values)

$$= (se^t)[3(se^t)^2 - 6(se^t)(se^{-t})^3] + (se^{-t})[9(se^t)^2(se^{-t})^2] = (se^t)[3s^2e^{2t} - 6s^2e^{t-3t}] + (se^{-t})[9s^4] = 3s^3e^{3t} - 6s^3e^{-t} + 9s^5e^{-t} = 3s^3[e^{3t} - 2e^{-t} + 3s^2e^{-t}]$$

Question 9.

$W(x,y,z)=xy+yz+zx, x=u-v, y=uv, z=u+v, u,v \in \mathbb{R}$. Find $\partial w \partial u, \partial w \partial v$ them at $(12,1)$

Solution:

$$W(x,y,z)=xy+yz+zx; x=u-v; y=uv; z=u+v$$

(i)

$$\partial w \partial u = \partial w \partial x \partial x \partial u + \partial w \partial y \partial y \partial u + \partial w \partial z \partial z \partial u = (y+z)(1) + (x+z)(v) + (x+y)(1) = y+z + (x+z)v + x+y$$

(Replace x&y values)

$$= uv + u + v + (u-v+u+v)v + (u-v+uv) = uv + u + v + 2uv + u - v + uv = 4uv + 2u \quad \partial w \partial u = 2u(1+2v)$$

(ii)

$$\partial w \partial v = \partial w \partial x \partial x \partial v + \partial w \partial y \partial y \partial v + \partial w \partial z \partial z \partial v = (y+z)(-1) + (x+z)(u) + (x+y)(1) = -y-z + (x+z)u + x+y = -z + (x+z)u + x = -u-v + (u-v+u+v)u + u-v = -2v + 2u^2$$

$$\partial w \partial v = 2[u^2 - v] \quad \partial w \partial u = 2(12)[1+2(1)] = 3 \quad \partial w \partial v = 2[14-1] = 2[-34] = -32$$

Ex 8.7

Question 1.

In each of the following cases, determine whether the following function is homogeneous or not. If it is so, find the degree.

(i) $f(x,y)=x^2y+6x^3+7$

(ii) $h(x,y)=6x^2y^3-\pi y^5+9x^4y^{2020}x^2+2019y^2$

(iii) $g(x,y,z)=\sqrt{3x^2+5y^2+z^2}4x+7y$

(iv) $U(x,y,z)=xy+\sin(y^2-2zzxy)$

Solution:

(i)

$$f(x,y)=x^2y+6x^3+7 \quad f(tx,ty)=(tx)^2(ty)+6(tx)^3+7 \quad f(tx,ty)=t^3(x^2y)+t^3(6x^3)+7$$

It is not a homogeneous function

(ii)

$$h(x,y)=6x^2y^3-\pi y^5+9x^4y^{2020}x^2+2019y^2 \quad h(tx,ty)=6(tx)^2(ty)^3-\pi(ty)^5+9(tx)^4(ty)^{2020}(tx)^2+2019(ty)^2$$
$$h(tx,ty)=t^5[6x^2y^3-\pi y^5+9x^4y^{2020}x^2+2019y^2] \quad h(tx,ty)=t^5h(x,y)$$

$\therefore$ It is a homogeneous function with degree 5.

(iii) $g(x,y,z)=\sqrt{3x^2+5y^2+z^2}4x+7y$

$$g(tx, ty, tz) = \sqrt{3t^2x^2 + 5t^2y^2 + t^2z^2} [4tx + 7ty] = \sqrt{t^2} [t \sqrt{3x^2 + 5y^2 + z^2} 4x + 7y] g(tx, ty, tz) = t^0 [g(x, y, z)]$$

$\therefore$ It is homogeneous function of degree 0 .

(iv)

$$U(x, y, z) = xy + \sin(y^2 - 2z^2) \partial_y U(tx, ty, tz) = (tx)(ty) + \sin(t^2y^2 - 2t^2z^2(tx)(ty)) = t^2(xy) + \sin(y^2 - 2z^2xy)$$

$\therefore$ It is not a homogeneous function

Education

Question 2.

Prove that $f(x, y) = x^3 - 2x^2y + 3xy^2 + y^3$ is homogeneous; what is the degree? Verify Euler's Theorem for f .

Solution:

$$f(x, y) = x^3 - 2x^2y + 3xy^2 + y^3 \quad f(tx, ty) = t^3x^3 - 2(t^2x^2)(ty) + 3(tx)(t^2y^2) + t^3y^3 = t^3[x^3 - 2x^2y + 3xy^2 + y^3] = t^3f(x, y)$$

$$f(tx, ty) = t^3f(x, y)$$

' f ' is a homogeneous function of degree 3. By Euler's theorem, we have

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 3f \quad \text{LHS} = x \frac{\partial}{\partial x} (x^3 - 2x^2y + 3xy^2 + y^3) + y \frac{\partial}{\partial y} (x^3 - 2x^2y + 3xy^2 + y^3) = x[3x^2 - 4xy + 3y^2] + y[-2x^2 + 6xy + 3y^2] = 3x^3 - 4x^2y + 3xy^2 - 2x^2y + 6xy^2 + 3y^3 = 3x^3 - 6x^2y + 9xy^2 + 3y^3 = 3[x^3 - 2x^2y + 3xy^2 + y^3] = 3f = \text{RHS}$$

$\therefore$ Euler's Theorem verified

Question 3.

Prove that $g(x, y) = x \log(yx)$ is homogeneous; what is the degree? Verify Euler's Theorem for g .

Solution:

$$g(x, y) = x \log(yx) \quad g(tx, ty) = (tx) \log(tytx) = tx \log(yx) = tg(x, y)$$

$\therefore$ ' g ' is a homogeneous function of degree 1. By Euler's theorem, we have

$$x \frac{\partial g}{\partial x} + y \frac{\partial g}{\partial y} = g \quad \text{LHS} = x[x(xy \cdot -yx^2) + \log(yx)] + y[x(xy \cdot 1x)] = x[-1 + \log(yx)] + x = -x + x \log(yx) + x = x \log(yx) = \text{RHS} \therefore \text{Euler's Theorem verified}$$

Question 4.

If $u(x, y) = x^2 + y^2 \sqrt{x+y}$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 32u$

Solution:

$$u(x, y) = x^2 + y^2 \sqrt{x+y} \quad u(tx, ty) = t^2x^2 + t^2y^2 \sqrt{tx+ty} = t^2 \sqrt{t} [x^2 + y^2 \sqrt{x+y}] u(tx, ty) = t^{3/2} u(x, y)$$

' u ' is a homogeneous function of degree $3/2$. By Euler's theorem, we have

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 32u$$

Question 5.

If $v(x, y) = \log(x^2 + y^2x + y)$, prove that $x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 1$

Solution:

$$v(x, y) = \log(x^2 + y^2x + y)$$

$$\text{Let } f = e^v = x^2 + y^2x + y$$

$\therefore$ ' f ' is a homogeneous function of degree 1. By Euler's theorem, we have

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf$$

(but $f = e^v$ and $n = 1$)

$$x \frac{\partial}{\partial x} (e^v) + y \frac{\partial}{\partial y} (e^v) = e^v x \frac{\partial v}{\partial x} + y e^v \frac{\partial v}{\partial y} = e^v x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 1$$

Question 6.

If $w(x, y, z) = \log(5x^3y^4 + 7y^2xz^4 - 75y^3z^4x^2 + y^2)$, find $x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} + z \frac{\partial w}{\partial z}$

Solution:

$$w(x, y, z) = \log[5x^3y^4 + 7y^2xz^4 - 75y^3z^4x^2 + y^2]$$

$$\text{Let } f = e^w = 5x^3y^4 + 7y^2xz^4 - 75y^3z^4x^2 + y^2$$

'f' is a homogeneous function of degree 5. By Euler's theorem, we have

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = nf$$

(But $n=5; f=e_w$)

$$x\frac{\partial}{\partial x}(e_w) + y\frac{\partial}{\partial y}(e_w)$$

$$+ z\frac{\partial}{\partial z}(e_w) = 5e_w e_w [x\frac{\partial w}{\partial x} + y\frac{\partial w}{\partial y} + z\frac{\partial w}{\partial z}] = 5e_w x\frac{\partial w}{\partial x} + y\frac{\partial w}{\partial y} + z\frac{\partial w}{\partial z} = 5$$

Ex 8.8

Choose the correct or the most suitable answer from the given four alternatives.

Question 1.

A circular template has a radius of 10 cm. The measurement of radius has an approximate error of 0.02 cm. Then the percentage error in calculating area of this template is

- (a) 0.2%
- (b) 0.4%
- (c) 0.04%
- (d) 0.08%

Solution:

- (b) 0.4%

Hint:

$$r=10 \text{ cm}, dr=0.02 \text{ cm} \quad A=\pi r^2 \quad dA=2\pi r dr \quad \frac{dA}{A} \times 100 = \frac{2\pi r dr}{\pi r^2} \times 100 = 2 \frac{dr}{r} \times 100 = 2 \times 100 \times \frac{0.02}{10} = 0.4\%$$

Question 2.

The percentage error of fifth root of 31 is approximately how many times the percentage error in 31 ?

- (a) 131
- (b) 15
- (c) 5
- (d) 31

Solution:

15

Hint:

We know that the percentage error in the n th root of a number is approximately $\frac{1}{n}$ times the percentage error in the number.

so 15

Question 3.

If $u(x,y)=e^{x^2+y^2}$, then $\frac{\partial u}{\partial x}$ is equal to

- (a) $e^{x^2+y^2}$
- (b) $2xu$
- (c) x^2u
- (d) y^2u

Solution:

- (b) $2xu$

Hint:

$$\frac{\partial u}{\partial x} = 2xe^{x^2+y^2} = 2xu$$

Question 4.

If $v(x,y)=\log(e^x+e^y)$, then $\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y}$ is equal to

- (a) e^x+e^y
- (b) $1e^{x+y}$
- (c) 2
- (d) 1

Solution:

- (d) 1

Hint:

$$\partial v \partial x = e_x e_x + e_y; \partial v \partial y = e_y e_x + e_y \partial v \partial x + \partial v \partial y = e_x + e_y e_x + e_y = 1$$

Question 5.

If $w(x,y) = xy, x > 0$, then $\partial w \partial x$ is equal to

- (a) $xy \log x$
- (b) $y \log x$
- (c) $y x^{y-1}$
- (d) $x \log y$

Solution:

- (c) $y x^{y-1}$

Hint:

$$w(x,y) = xy$$

$$\partial w \partial x = y x^{y-1}$$

Question 6.

If $f(x,y) = e^{xy}$, then $\partial^2 f \partial x \partial y$ is equal to

- (a) $xy e^{xy}$
- (b) $(1+xy) e^{xy}$
- (c) $(1+y) e^{xy}$
- (d) $(1+x) e^{xy}$

Solution:

- (b) $(1+xy) e^{xy}$

Hint:

$$f(x,y) = e^{xy} \partial f \partial y = x e^{xy} \partial^2 f \partial x \partial y = x [y e^{xy} + e^{xy}] = e^{xy} [xy + 1]$$

Question 7.

If we measure the side of a cube to be 4 cm with an error of 0.1 cm, then the error in our calculation of the volume is

- (a) 0.4 cu.cm
- (b) 0.45 cu.cm
- (c) 2 cu.cm
- (d) 4.8 cu.cm

Solution:

- (d) 4.8 cu.cm

Hint.

$$a = 4 \text{ cm}, da = 0.1 \text{ cm}, v = a^3, dv = 3a^2 da = 3(4)^2(0.1) = 4.8 \text{ cu.cm}$$

Question 8.

The change in the surface area $S = 6x^2$ of a cube when the edge length varies from x_0 to $x_0 + dx$ is

- (a) $12x_0 + dx$
- (b) $12x_0 dx$
- (c) $6x_0 dx$
- (d) $6x_0 + dx$

Solution:

- (b) $12x_0 dx$

Question 9.

the approximate change in the volume: V of a cube of side x metres caused by increasing the side by 1% is

- (a) $0.3x dx m^3$
- (b) $0.03x m^3$
- (c) $0.03x^2 m^3$
- (d) $0.03x^3 m^3$

Solution:

- (d) $0.03x^3 m^3$

Hint:

Let the side of the cube be x units

$$V = x^3$$

$$\text{when } dx = 0.01x$$

$$dV = 3x^2 dx = 3x^2(0.01x) = 0.03x^3 m^3$$

Question 10.

If $g(x, y) = 3x^2 - 5y + 2y^2$, $x(t) = e^t$ and $y(t) = \cos t$, then dg/dt is equal to

- (a) $6e^{2t} + 5\sin t - 4\cos t \sin t$
- (b) $6e^{2t} - 5\sin t + 4\cos t \sin t$
- (c) $3e^{2t} + 5\sin t + 4\cos t \sin t$
- (d) $3e^{2t} - 5\sin t + 4\cos t \sin t$

Solution:

- (a) $6e^{2t} + 5\sin t - 4\cos t \sin t$

Question 11.

If $f(x) = x^{x+1}$ then its differential is given by

- (a) $-1(x+1)^2 dx$
- (b) $1(x+1)^2 dx$
- (c) $1x+1 dx$
- (d) $-1x+1 dx$

Solution:

- (b) $1(x+1)^2 dx$

Hint:

$$df = (x+1)(1) - x(1+0)(x+1)^2 dx = 1(x+1)^2 dx$$

Question 12.

If $u(x, y) = x^2 + 3xy + y - 2019$, then $\partial u / \partial x$ at $(4, -5)$ is equal to

- (a) -4
- (b) -3
- (c) -7
- (d) 13

Solution:

- (c) -7

Hint:

$$\partial u / \partial x = 2x + 3y \quad (\partial u / \partial x)_{(4, -5)} = 8 - 15 = -7$$

Question 13.

Linear approximation for $g(x) = \cos x$ at $x = \pi/2$ is

- (a) $x + \pi/2$
- (b) $-x + \pi/2$
- (c) $x - \pi/2$
- (d) $-x - \pi/2$

Solution:

(b) $-x + \pi^2$

Hint:

$g(x) = \cos x; x_0 = \pi^2 g'(x) = -\sin x g(\pi^2) = \cos(\pi^2) = 0 g'(\pi^2) = -\sin(\pi^2) = -1$ Linear approximation $L(x) = g(x_0) + g'(x_0)(x - x_0) = 0 + (-1)(x - \pi^2) = -x + \pi^2$

Question 14.

If $w(x, y, z) = x^2(y - z) + y^2(z - x) + z^2(x - y)$, then (a) $xy + yz + zx$

(a) $xy + yz + zx$

(b) $x(y + z)$

(c) $y(z + x)$

(d) 0

Solution:

(d) 0

Hint:

$$\partial w / \partial x = 2x(y - z)$$

$$-y^2 + z^2 = 2xy - 2xz - y^2 + z^2 \partial w / \partial y = x^2 + 2yz - z^2 \partial w / \partial z = -x^2 + y^2 + 2zx - 2zy \partial w / \partial x + \partial w / \partial y + \partial w / \partial z = 0$$

Question 15.

If $f(x, y, z) = xy + yz + zx$, then $f_x - f_z$ is equal to

(a) $z - x$

(b) $y - z$

(c) $x - z$

(d) $y - x$

Solution:

(a) $z - x$

Hint:

$$f_x = y + z f_z = y + x f_x - f_z = y + z - y - x = z - x$$